Uncertainty and Unruhigkeit:

A Galois Connection

Companion paper to: Syntax, Geometry, and Measurement

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March 2026

Epigraph

"Show and tell are a conjugate pair: an actor issues orders, and a universe demonstrates the response."

Abstract

*We develop a non-ecological thermodynamics parallel to the non-ecological mechanics of [Paper 1]. Temperature is treated as a primitive scalar field T on a Level 3 spacetime, satisfying the field equation Box(T) + (1/3)RT = 0, derived from the free-streaming Liouville equation on the geodesic flow without any particle content. The central result is that curvature and temperature are conjugate in the Galois connection sense: their level surfaces coincide, causal isolation boundaries are isothermal surfaces, and curvature singularities are temperature point sources. The Unruh effect is reinterpreted as the canonical Level 1 thermodynamic result — derivable from two geometric facts alone, requiring no quantum field theory. The equivalence principle is a tautology. The Schwarzschild temperature field is computed explicitly, yielding the Hawking temperature T_H = 1/4pi*r_s as a pure geometric result. The non-ecological Tolman relation r_eff * T = 1/4pi is derived and shown to be identical to the thermodynamic uncertainty relation S * T^2 = 1/16pi. The framework contains only c and pi; the appearance of hbar, G, k_B, and e in standard formulae is identified as ecological bookkeeping.*

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1. Introduction and Motivation

This paper is a companion to [Paper 1], which developed a unified foundational programme separating structural mathematics from ecological data across logic, geometry, and physics. Here we extract and develop the thermodynamic thread of that programme into a focused result: a non-ecological thermodynamics in which temperature is a primitive geometric scalar field, and curvature and temperature are conjugate in the Galois connection sense.

1.1 SR's contaminated origin

Special relativity is conventionally presented as foundational. But Einstein's 1905 paper was titled 'On the Electrodynamics of Moving Bodies' — its motivation was entirely electromagnetic, and the postulate about the speed of light is at root a postulate about the speed of electromagnetic waves. SR's foundational structure is contaminated at birth by the ecology of electromagnetism. The Level 0-1 framework of [Paper 1] achieves the necessary disentanglement.

1.2 The Majumdar-Papapetrou warning

The Majumdar-Papapetrou solutions describe multiple extremal black holes in static equilibrium, with gravitational attraction exactly balanced by electromagnetic repulsion. They are elegant mathematics but deeply ecological: charge is a concept from the standard model, and the balance condition ties the geometry to a specific matter coupling. We are chargeless by construction. These solutions are not available to us, and their elegance should not tempt us to reintroduce electromagnetism through the back door.

1.3 The equivalence principle as tautology

In our framework there is no mass and no force — only geodesic motion in a curved causal structure. The equivalence principle, asserting that gravitational and inertial mass are equal, has no content to assert. Curvature is the deviation from straight causal propagation — acceleration and curvature are definitionally related. The principle dissolves into a tautology, which is not a failure but a success: we have found the right level of description.

1.4 Carathéodory and natural units

Carathéodory's 1909 axiomatisation of thermodynamics is our direct predecessor in spirit: no Carnot cycles, no ideal gases, no ecological substance — just the geometry of state space and the inaccessibility condition. We carry this further into the relativistic setting.

Natural units in our framework admit only c and pi. The constants hbar, k_B, G, and e are ecological bookkeeping — conversion factors between geometric and physical units. Their appearance in any formula is an immediate signal of ecological contamination. The Hawking temperature formula T_H = hbar*c^3 / 8*pi*G*M*k_B fails this diagnostic; its non-ecological residue is T_H = 1/4*pi*r_s, which passes it.

2. Particle-Free Classical Thermodynamics

The central simplification of the non-ecological setting is the collapse of the adiabatic/isothermal distinction. In standard thermodynamics these are genuinely different: an adiabatic process has no heat exchange, requiring a working substance; an isothermal process is at constant temperature, requiring thermal equilibrium mediated by collisions. Both distinctions require particles. In a particle-free setting, both collapse to a single geometric condition: **causal isolation**. There is no mechanism of energy exchange other than causal propagation. A region is adiabatically isolated if and only if it is causally isolated. And a causally isolated region has a well-defined boundary temperature — constant on its boundary by the isothermal condition. Therefore: adiabat = isotherm = causal isolation boundary. This is a theorem, not an approximation.

There are no spin-statistics complications: the temperature field is classical and geometric. Bose-Einstein and Fermi-Dirac distributions are ecological additions that arise only when particles are reintroduced.

2.1 The four laws, recast

Zeroth law. Causal contact implies equal boundary temperature. Two regions in causal contact share an isothermal boundary — geometric, not statistical.

First law. Conservation follows from the Bianchi identities. No heat, no work in the ecological sense — only geometric exchange encoded in the curvature field.

Second law. Causal order is irreversible. Entropy is non-decreasing along causal geodesics. This is Carathéodory's inaccessibility condition, now identified exactly with causal inaccessibility.

Third law. T = 0 at causally flat regions where curvature vanishes and geodesics do not focus.

2.2 The noise floor as event horizon

The noise floor — the informational boundary beneath which no structure is recoverable — is not merely analogous to the event horizon. It is the event horizon, expressed in thermodynamic language. The event horizon is the causal boundary beyond which no signal returns; the noise floor is the informational boundary beneath which no structure is recoverable. They are the same boundary seen from the causal and thermodynamic sides of the adjunction respectively.

2.3 The non-ecological tensor inventory

Energy is an ecological form of entropy — the geometric entropy field weighted by the ecological temperature, coupling constants, and particle content. In our formalism, energy does not exist as a primitive, just as momentum and force do not exist. This has immediate consequences for the tensor inventory.

The standard stress-energy tensor T_mu_nu — named for energy and momentum, both ecological — is more honestly called the **entropy-flux tensor** Sigma_mu_nu: it encodes the distribution of entropy flux and causal influence across the manifold, without reference to energy or momentum. When ecology is reintroduced and particles appear, Sigma_mu_nu acquires the interpretation of energy-momentum density — but that is an ecological reading of a geometric object.

The Maxwell tensor F_mu_nu does not appear in this formalism. We are chargeless by construction. Its absence is not an omission but a feature — electromagnetism is ecology, and the entire electromagnetic sector of the standard model is absent. The cone structure is not the speed of light but the geometric feature of the causal order that light happens to saturate in our particular world.

The non-ecological tensor inventory, with honest names:

$$g_{\mu\nu} \text{ — causal structure tensor (cone geometry, Level 1)}$$

$$R_{\mu\nu\rho\sigma} \text{ — curvature tensor (primitive field, Level 3)}$$

$$R_{\mu\nu} \text{ — geodesic focusing tensor (Raychaudhuri source)}$$

$$G_{\mu\nu} \text{ — geometric divergence tensor (Bianchi contraction)}$$

$$\Sigma_{\mu\nu} \text{ — entropy-flux tensor (replaces } T_{\mu\nu}\text{)}$$

No $F_{\mu\nu}$. No momentum. No force. Energy is an ecological form of entropy, and the tensor inventory makes this precise.

3. The Level Hierarchy for Thermodynamics

Level 0 — Entropy as causal path counting

Entropy is the logarithm of the number of causally accessible microstates — the count of distinct causal paths available to the system. No dynamics, no Hamiltonian, no phase space measure beyond what the causal order provides. The arrow of time follows immediately: causal order is irreversible, so entropy is non-decreasing.

Level 1 — Temperature from cone structure

The canonical Level 1 thermodynamic result is the Unruh effect, properly understood. An accelerating observer sits on a non-trivial isothermal surface of the temperature field — their Rindler horizon is their personal noise floor, the boundary of their causal accessibility. The temperature they perceive is the monopole coefficient of the temperature point source at their causal horizon: T proportional to a/c^2. No particle physics. No hbar. Acceleration is heating without chemistry.

Level 2 — Tolman-Ehrenfest

The first ecological level. The Minkowski metric enters, and with it the standard Tolman-Ehrenfest relation for temperature in flat spacetime with matter. This is SR thermodynamics — a special ecological case of what follows.

Level 3 — Temperature as a scalar field

At Level 3, before any ecology, temperature is simply a scalar field $T : M \to R^+$ on the spacetime manifold. It is a geometric object, as primitive as the curvature tensor. The Tolman relation $T\sqrt{g_{00}} = \text{const}$ is visible here as a hint of the conjugacy — temperature and metric component are inversely related in a precise way. In the non-ecological limit this becomes T and R directly.

Note: as in [Paper 1], it is illuminating to proceed backward from Level 3 — the most general case reveals what the special cases were concealing. SR thermodynamics conceals the scalar field nature of T by fixing the metric; our framework reveals it.

4. The Temperature Field Equation

4.1 From Raychaudhuri and Bianchi

Let T be a scalar field with isothermal surfaces $\Sigma_c = \{T = c\}$ identified with causal isolation boundaries. Let the congruence of curves within Σ_c have tangent u^μ, with vorticity $\omega_{\mu\nu} = 0$ (hypersurface orthogonality). At a non-expanding horizon, $d\theta/d\tau = 0$ and $\theta = 0$. The Raychaudhuri equation gives:

$$R_{\mu\nu} u^\mu u^\nu = -\sigma_{\mu\nu} \sigma^{\mu\nu} \leq 0$$

The contracted Bianchi identity gives:

$$\nabla^\mu R_{\mu\nu} = (1/2) \nabla_\nu R$$

Projecting onto the temperature gradient $n^\mu = \nabla^\mu T / |\nabla T|$ yields the integrability condition:

$$\nabla_\mu T \propto \nabla_\mu R$$

The gradients of T and R are aligned. Their level surfaces coincide. The temperature field and the Ricci scalar field foliate the manifold in the same way. This is the geometric content of their conjugacy.

4.2 From the free-streaming Liouville equation

The non-ecological Boltzmann equation — the free-streaming Liouville equation on the geodesic flow — is:

$$u^\mu \nabla_\mu f = 0$$

No particles, no collisions, no ecology. Identifying temperature as the zeroth moment of the geodesic distribution:

$$T \propto \rho = \int f \, d\Omega$$

Taking the zeroth moment of the Liouville equation and commuting covariant derivatives — which generates Ricci tensor terms via the Ricci identity — yields the field equation:

$$\boldsymbol{\Box T + (1/3)\, R\, T = 0}$$

This is the non-ecological temperature field equation. It is a wave equation with curvature-dependent effective mass $m^2 = R/3$. It contains only c and pi. In flat spacetime $R = 0$ and the equation reduces to $\Box(T) = 0$ — temperature is a free massless scalar, consistent with Level 2.

The factor 1/3 arises from the trace over three spatial dimensions of the geodesic congruence. Compare with the conformally coupled scalar field $\Box(\phi) + (1/6)R\phi = 0$ — our coefficient is twice the conformal value, suggesting T is not conformally invariant. We conjecture this non-invariance is the geometric expression of the second law.

4.3 The entropy field

Setting S proportional to $\log(T)$, the entropy field satisfies:

$$\Box S + |\nabla S|^2 + (1/3)R = 0$$

This is a Hamilton-Jacobi equation — entropy is the action of the thermodynamic geometry. The analogy with WKB and the semiclassical limit of quantum gravity is striking and probably not accidental.

5. Curvature and Temperature as a Conjugate Pair

5.1 The Galois connection

The Galois connection between T and R, in the language of [Paper 1]:

not side: causal isolation boundaries — level surfaces of T, freely determined by the causal order, requiring no ecological data.

neg side: curvature sources — singularities sourcing R, requiring ecological specification to particularise.

The Galois axioms: nothing is its own curvature source; the deep interior of an isolation boundary lies within it. These follow from the alignment condition $\nabla_\mu T$ proportional to $\nabla_\mu R$.

5.2 Curvature singularities as temperature point sources

Curvature singularities are distributional sources in the field equation. In the exterior, R = 0, so Box(T) = 0 — the temperature field is sourced only by the singularity. Just as the Newtonian potential satisfies nabla^2(phi) = 0 in the exterior with a point source at the origin, T satisfies Box(T) = 0 in the exterior with a distributional source at the singularity.

The Hawking temperature T_H is the monopole coefficient of the temperature field around the singularity — the leading term in the multipole expansion of T. Higher multipole moments correspond to deviations from spherical symmetry: rotation (Kerr) generates a dipole moment, deformation generates quadrupole, and so on. This structure has not previously been examined from this geometric angle.

5.3 The non-ecological Unruh effect

The Unruh effect follows immediately from two geometric facts:

Fact 1. Level surfaces of T and R coincide — the boundary of a region of non-zero curvature is simultaneously an isothermal surface.

Fact 2. Curvature singularities are temperature point sources — localised divergences in both fields simultaneously.

An accelerating observer is, by the equivalence principle-as-tautology, locally equivalent to an observer in a curved region. They therefore sit on a non-trivial isothermal surface and perceive the monopole temperature of the effective curvature source at their Rindler horizon. No quantum fields. No Bogoliubov transformations. No hbar.

The non-ecological Unruh formula is:

$$T = a / 2\pi c^2$$

containing only c and pi. The standard formula T = hbar*a / 2*pi*c*k_B is the ecological bookkeeping version — hbar and k_B are unit conversion factors between geometric and physical descriptions.

5.4 The Schwarzschild temperature field

In the Schwarzschild exterior, R = 0, so the field equation reduces to Box(T) = 0. For a static, spherically symmetric T = T(r), this gives:

$$d/dr \, [\, (r^2 - r_s \, r) \, dT/dr \,] = 0$$

Integrating twice with boundary condition $T \to 0$ as $r \to \infty$:

$$\mathbf{T(r) = (1/4\pi) \log(1 - r_s/r)}$$

This is the exact non-ecological Schwarzschild temperature field. Its features:

— Monopole tail: T ~ –r_s/4pi*r at large r, falling off as 1/r

— Logarithmic divergence at r = r_s: the horizon is an extended temperature source, not a point source

— Hawking temperature as horizon source strength:

$$T_H = 1 / 4\pi \, r_s$$

containing only pi and the geometric length r_s. No hbar, G, or k_B.

5.5 The non-ecological Tolman relation

Temperature is proportional to the square root of the surface curvature of the isothermal surface:

$$T_H = (1/4\pi) \sqrt{\kappa\_surface} = 1/4\pi \, r_s$$

since the Gaussian curvature of the horizon 2-sphere is kappa = 1/r_s^2. Entropy is proportional to the area:

$$S = A/4 = \pi \, r_s^2$$

The effective radius of the isothermal surface is r_eff = sqrt(A/4pi) = r_s. Therefore:

$$\mathbf{r\_eff \cdot T = 1 / 4\pi}$$

This is the **non-ecological Tolman relation** — temperature times effective radius is a pure geometric constant. It contains only pi, passes the ecological diagnostic, and reduces to Tolman's original result T*sqrt(g_00) = const when the metric is reinstated ecologically.

5.6 The thermodynamic uncertainty relation

From S = pi*r_s^2 and T = 1/4pi*r_s:

$$\mathbf{S \cdot T^2 = 1 / 16\pi}$$

This is the **thermodynamic uncertainty relation** — a product of conjugate quantities equalling a pure geometric constant. It has the structure of the Heisenberg relation Delta(x)*Delta(p) >= hbar/2, but here the constant is 1/16pi, purely geometric, and the relation is an equality rather than an inequality.

The equality holds for the Schwarzschild solution — the maximally coherent, spherically symmetric state. For Kerr (rotation) the equality becomes an inequality:

$$S \cdot T^2 \geq 1 / 16\pi$$

with equality only at zero rotation. This is the gravitational uncertainty principle.

Most strikingly: the non-ecological Tolman relation and the thermodynamic uncertainty relation are **the same equation**. Both state that r_eff * T = 1/4pi. The uncertainty relation is not additional structure imposed on the geometry — it is already contained in the Tolman relation, properly understood in the non-ecological setting.

6. The Unruh Effect Reinterpreted [to be expanded]

Standard derivation via Bogoliubov transformations and Rindler wedge — ecological loading identified explicitly. All instances of hbar and k_B flagged as unit conversion. The non-ecological residue: T proportional to a/c^2, derivable from Facts 1 and 2 of Section 5.3 alone.

The Rindler horizon as Level 1 causal feature — the personal noise floor of the accelerating observer. The thermal bath as causal complement of the Rindler wedge. The universe showing what is being attempted.

Comparison with Hawking radiation: Hawking is the static version of Unruh. Both are Level 1 results contaminated by ecological presentation. The logarithmic divergence of T(r) at r = r_s is the Unruh effect made visible in the temperature field.

Acceleration as heating without chemistry: the purest thermodynamic process, stripped of every ecological mechanism.

7. The Equivalence Principle as Tautology [to be expanded]

Standard statement and its ecological loading. In our framework: no mass, no force, only geodesic deviation. Curvature is the deviation from straight causal propagation — acceleration and curvature are definitionally related.

The dissolution/sharpening diagnostic: equivalence principle dissolves; Unruh effect sharpens. What the equivalence principle was pointing at: the Level 1 identification of acceleration with cone structure.

The tautology is not a failure but a success — it locates the correct level of description.

8. Uncertainty, Temperature, and the Full Conjugate Table [to be expanded]

Revisiting Paper 1's relativistic uncertainty: x_B = Tc/2, collapse structure, conjugate variables in a massless universe.

The full conjugate table:

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position / velocity (Paper 1, Level 1)
curvature / temperature (Paper 2, Level 3)
observation / instruction (logical level)
uncertainty / cost (Section 7 of Paper 1)
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Vaccarino-Barnett: all four pairs are valid accounting numeraires. The uncertainty relation S*T^2 = 1/16pi as the Level 3 entry in this table. Kerr as the next step: vorticity, angular momentum, dipole moment of T. The inequality S*T^2 >= 1/16pi for Kerr.

9. Synthesis and Open Questions [to be expanded]

Non-ecological mechanics and non-ecological thermodynamics as two faces of the causal skeleton. The Boltzmann/Liouville connection: Box(T) + (1/3)RT = 0 as zeroth moment of free-streaming geodesic flow.

The factor 1/3 vs 1/6: geometric significance, relationship to conformal invariance. Conjecture: non-invariance is the geometric content of the second law.

Open questions within this paper: (1) Precise uncertainty relation for Kerr — functional form of S*T^2 >= 1/16pi with angular momentum J: conjecture S*T^2 >= 1/16pi*(1 + J^2/M^2), equality at J=0. (2) Whether Box(T) + (1/3)RT = 0 follows from a variational principle without an action — from Bianchi identities alone. (3) The relationship between the Hamilton-Jacobi structure of S and the semiclassical limit of quantum gravity. (4) Whether quantisation arises naturally at Level 3, or requires a new level. (5) The Kerr multipole expansion of T: dipole from rotation, quadrupole from deformation — vorticity omega_mu_nu non-zero modifies the Raychaudhuri equation and lifts the equality.

Unruhigkeit is conjugate to Unruheffekt. The title states the theorem. Acceleration (restlessness, the actor issuing the instruction) is conjugate to the Unruh effect (the universe demonstrating the response). Their product is the geometric constant 1/4pi. The Unruhe — the balance wheel — regulates the exchange.

Forward Reference: Paper 3

The thermodynamic framework of this paper extends naturally to the topology of the geodesic flow — the subject of the companion paper: Vorticity, Helicity, and Curvature Rotons: Topological Thermodynamics of the Geodesic Flow [Paper 3].

The central analogy: vorticity omega = curl(v) in fluids corresponds to curvature R_mu_nu_rho_sigma in the geodesic flow. Helicity H = integral(v.omega dV) is the fluid analogue of gravitational entropy — a topological invariant counting the linking of vortex lines, related to the Gauss linking integral and Stokes theorem exactly as the Bianchi identities relate local curvature to global causal structure.

Curvature rotons: the dispersion relation for the temperature field Box(T) + (1/3)RT = 0 in a region of approximately constant R gives omega^2 = k^2 + R/3 — a massive scalar dispersion with a mass gap at k=0 of omega_min = sqrt(R/3). This is the roton signature. Excitations near this minimum are curvature rotons — collective excitations of the curvature field at the characteristic focusing length l_f ~ R^(-1/2) set by the Raychaudhuri equation.

Curvature rotons may correspond to new oscillatory vacuum solutions of the Bianchi-constrained curvature dynamics — neither singular nor flat but periodic, at the characteristic scale l_f. In the ecological presentation these would be quasi-normal modes of the gravitational field derived from geometric first principles rather than perturbation theory. The crossover between roton-like and quasiparticle-like behaviour is a topological transition governed by the Raychaudhuri equation.

Turbulence as thermodynamic equilibrium: the Kolmogorov cascade is the renormalisation group flow on the topological configuration space of the vorticity field. The k^(-5/3) spectrum is the fixed point of the thermodynamic uncertainty relation for the vortex configuration space. Turbulence is the receipt for driving a fluid far from its geometric ground state — the universe showing what is being attempted.

Acknowledgements. This paper was developed through an extended dialogue between the authors, as a companion to [Paper 1]. The human author identified the key conjecture — curvature is conjugate to temperature — on a walk. The Platonic tradition is again thanked for the conversational form.